

closure_evidence_20260923

BOB-167 closure evidence — 2026-09-23

Defect

Two Server-Sent-Events routes in `download-proxy/src/api/routes.py` — the same expensive resource-holding shape — were asymmetrically classed: `/search/stream/{search_id}` carried `@_rl("sse_stream")`, `/theme/stream` carried no limiter decorator and fell through to the 120/min application default (24x looser).

Decision (coordinator, §11.4.101 — reversible, evidence-based)

`/theme/stream` belongs in the `sse_stream` class, matching its sibling route's existing shape exactly.

Fix

- `@_rl("sse_stream")` added to `/theme/stream` (`routes.py:161`).
- New general guard `scripts/pre_build/check_cm_sse_route_rate_limit_classed.sh` (+ AST-based `sse_route_rate_limit_class_analyzer.py`) enumerates every SSE-shaped route (both the inline `StreamingResponse(media_type=...)` idiom and the shared `create_streaming_response(...)` helper) and FAILs the build if any carries no explicit rate-limit class — so a future third SSE route cannot silently repeat this gap.
- Two new test files (`tests/unit/api_layer/test_bob167_sse_rate_limit_classification.py`, `tests/unit/`

api_layer/test_bob167_sse_route_rate_limit_gate.py, 10 tests total) covering both routes' classification, the general gate's golden-good/golden-bad fixtures (6 shapes including a non-SSE StreamingResponse negative control and a zero-SSE-route ERROR-not-silent-PASS guard), and a permanent in-memory paired-mutation regression test that mechanically strips the fix and asserts the gate re-fails.

Independent verification (coordinator, from clean shell)

Diff scope confirmed minimal

```
download-proxy/src/api/routes.py | 11 ++++++++
```

Plus 4 new files (gate wrapper, analyzer, 2 test files) — matches the implementing agent's own report exactly.

RED independently reproduced (via git stash, not trusted from report)

```
$ git stash push -m "..." -- download-proxy/src/api/routes.py
$ bash scripts/pre_build/check_cm_sse_route_rate_limit_classed.sh
FAIL(CM-SSE-ROUTE-RATE-LIMIT-CLASSED): 1 unclassified SSE-shaped
route(s) of 2 resolved.
$ .venv/bin/python -m pytest tests/unit/api_layer/
test_bob167_sse_rate_limit_classification.py -q
1 failed, 1 passed
$ git stash pop # byte-identical restore confirmed via
py_compile
```

GREEN independently confirmed (unit level)

```
$ bash scripts/pre_build/check_cm_sse_route_rate_limit_classed.sh
PASS(CM-SSE-ROUTE-RATE-LIMIT-CLASSED): all 2 SSE-shaped route(s)
carry an explicit rate-limit class.
$ .venv/bin/python -m pytest tests/unit/api_layer/
test_bob167_sse_rate_limit_classification.py tests/unit/api_layer/
test_bob167_sse_route_rate_limit_gate.py -q
10 passed
```

LIVE RUNTIME SIGNATURE — the strongest evidence tier (§11.4.108), beyond

what the implementing agent's own scope covered:

A pre-existing `xfail(strict=True)` test (`tests/scaling/test_scaling_envelope.py::TestRateLimitAdmissionEnvelope::test_sse_shaped_routes_serve_a_consistent_limit_class`), authored by the coordinator when BOB-167 was originally filed, was flagged by the implementing agent as an out-of-scope concern: its `strict=True` marker is deliberately load-bearing (its own comment: “the moment both SSE routes carry the same class this flips to XPASS and FAILS the run, forcing this marker's removal”). The coordinator investigated this rather than leaving it as a residual gap:

1. The merge service was live (`curl 200`) but still serving PRE-fix code — running the `xfail` test against it produced genuine XFAIL (not XPASS), confirming the source fix alone does not update a running container (the exact SOURCE→ARTIFACT→RUNTIME distinction §11.4.108 exists to catch).
2. Reloaded via the project's own sanctioned mechanism: `./start.sh --reload-python` — clears `__pycache__` inside `qbittorrent-proxy` and restarts the container (CLAUDE.md-documented path, not a raw `podman/docker` command).
3. Verified served content matches committed code (cache-bust guard): `podman exec qbittorrent-proxy grep -n '@_rl("sse_stream")' /config/download-proxy/src/api/routes.py` → both routes confirmed present in the running container.
4. Re-ran the `xfail` test:

```
FAILED ... [XPASS(strict)] BOB-167: two SSE routes, one rate-limit class...
```

Genuine `XPASS(strict)` — live, runtime proof the fix works against the real running service, not just unit-level mocks.

5. Removed the now-obsolete `@pytest.mark.xfail(...)` decorator (its own documented self-clearing condition, exactly as designed) and confirmed the test PASSES cleanly:

```
test_sse_shaped_routes_serve_a_consistent_limit_class PASSED
```

6. 3x determinism, live stack:

run 1: 1 passed in 0.24s
run 2: 1 passed in 0.26s
run 3: 1 passed in 0.26s

Honest boundary

Per the item's own criterion (d): no claim is made that 5/minute (or the previous 120/minute) is itself an adequate limit value for either route in this deployment — this closure is scoped purely to the classification symmetry gap.

Status

Fixed. Closed by coordinator after independent verification at BOTH the unit-test layer AND a live-runtime-signature layer beyond the implementing agent's own authorized scope.